

Ayush Ranjan

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LinkedIn — GitHub

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C, Dart, Rust, SQL

Frameworks & Libraries: React, Node.js, Flutter, Next.js, TensorFlow, PyTorch, scikit-learn

Databases: MySQL, MongoDB, Supabase, PostgreSQL, SQLite

Enterprise Applications: Oracle Argus (Currently Learning)

Tools & DevOps: Docker, Kubernetes, Git, REST APIs, CI/CD, Linux

Education

B.Tech in Computer Science & Engineering - KIIT University

2022 - 2026

Work Experience

Frontend Developer Intern - Tecosys

Jun 2024 - May 2025

- Engineered a fully responsive **LLM-integrated chat interface** using **React.js**, enhancing real-time communication and increasing user engagement by **30%**.
- Integrated **RESTful APIs** with optimized asynchronous data flow, improving system response speed by **25%** and maintaining **100% uptime**.
- Directed a **4-member development team**, coordinating tasks, reviewing pull requests, and ensuring on-schedule feature delivery.

Tech Intern - Yellow Slice

May 2025 - Dec 2025

- Designed and normalized the **Supabase database schema** for the Holivus app, reducing redundant records by **40%** and boosting data retrieval speed.
- Implemented modular **UI components** (Quiz Screen, Home Screen) in **Next.js** and **Flutter**, leading to a **30%** gain in application performance.

Key Projects

Sys Logger - Real-Time System Monitoring (Python, Flask, Next.js, Docker)

- Full-stack **distributed systems** dashboard: real-time **CPU, RAM, GPU** and network telemetry, live charts; **Flask** API and **Next.js** frontend (TypeScript, Tailwind CSS).
- Deployed at NIELIT Bhubaneswar; **Docker** containerized; configurable polling and health-check endpoints for fleet telemetry.

Krishi-Sahayogi - Smart Farming App (Flutter, MobileNetV2, Machine Learning)

- Developed an AI-based **machine learning disease detection model** achieving **98.3% accuracy** on **2,000+ test images**; **multi-language** recommendations for **50+ crops**.
- Full-stack mobile app: weather integration, live mandi data, marketplace, role-based dashboards; deployed in production.

Apex OS - Neural-Native Operating System (Rust, Systems Programming)

- Building a research-grade OS in **Rust** with **Reinforcement Learning-based CPU scheduling** and neural flight control (SAS) for swarm robotics; kernel-level world-model integration for predictive fairness.
- Implementing fail-secure hardening: capability-based security, TPM 2.0, verified boot, and tamper-evident audit chain.
- Designed **swarm hive-mind** architecture: local scheduling driven by collective system state for coordinated behavior across distributed nodes.

Leadership & Extracurricular

Team Lead, UI/UX - Microsoft Learn Student Ambassadors (MLSA) Club

Design strategy for 5+ platforms; **10+** technical events.